

MICHAEL CHADOLIAS

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Physics graduate specializing in astroparticle physics, with hands-on experience in neutrino physics, large-scale simulation pipelines, and machine learning applications in high-energy physics. Credited as a co-author on nine peer-reviewed KM3NeT publications for contributions to the collaboration. Currently seeking to pursue a PhD in particle physics.

RESEARCH INTERESTS

Neutrino physics, Particle & astroparticle physics, Statistical & Computational methods, Data science & AI

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg

Erlangen, Germany

M.Sc. in Physics (Specialisation: Astrophysics and astroparticle physics)

Apr 2022 – Feb 2025

Grade: 1.6/1.0 Equivalent grade: 90/100

Research: Active participation in the ANTARES/KM3NeT collaboration

Aristotle University of Thessaloniki (AUTH)

Thessaloniki, Greece

B.Sc. in Physics; Grade: 7.84/10.00

Oct 2018 – Feb 2022

Teaching Assistant: HM 131: Systems Reliability under Prof. Alkiviadis Hatzopoulos

Among the top 10% of students during my graduation period

RESEARCH EXPERIENCE

Erlangen Centre for Astroparticle Physics (ECAP)

Erlangen, Germany

Master Thesis Student

Aug 2023 – Oct 2024, Full-time

- Conducted the first feasibility study on tau neutrino appearance using ANTARES simulated data, evaluating the reconstruction quality of the events and providing the first estimate of the detector's capabilities.
- Developed a scalable pipeline to convert over 500k ANTARES Monte Carlo event files to TTree-based ROOT files, reducing pre-processing time by 20%, storage requirements by 35%, and enabling parallelised cluster execution (NHR@FAU & CC@Lyon).
- Analysed 200+ different oscillation scenarios with the SWIM software, accounting for the possible cut selection criteria, model parameters, and systematic uncertainties.
- Compared the tau appearance potential from ANTARES against KM3NeT/ORCA predicted sensitivities.

Erlangen Centre for Astroparticle Physics (ECAP)

Erlangen, Germany

Graduate Research Assistant

Nov 2022 – Apr 2023, Part-time

- Co-developed and tested modules within a Snakemake-based MC production pipeline for ORCA events.
- Performed testing and diagnostics of pipeline jobs on NHR@FAU cluster, reducing execution errors and improving workflow reproducibility.
- Achieved 25% CPU workload reduction by improving the SLURM profile, identifying inefficient parallelisation patterns, and optimising Snakemake rule resources.

National Centre of Scientific Research Demokritos (NCSR)

Athens, Greece

Undergraduate Research Assistant

Nov 2021 – Jan 2022, Internship

- Studied the reconstruction quality of simulated events focusing on the limited 6-string KM3NeT/ORCA detector configuration (ORCA6).
- Analysed ν_μ interaction topologies across key metrics such as energy, zenith angle, inelasticity, and distance from the detector, among others.
- Assisted in the optical module integration and component testing.

SpaceDot Team - AcubeSAT Project

Thessaloniki, Greece

Undergraduate Researcher & Sub-team Coordinator

Mar 2019 – Dec 2021, Part-time

- Led the trajectory subsystem and co-authored a proposal for ESA's "Fly Your Satellite" programme
- Modelled the space environment for the mission with the OMERE, and estimated the radiation levels during the entire duration of the mission with the FASTRAD software, with this work being credited as my bachelor's thesis.

RELEVANT COURSEWORK

Specialisation: Particle Physics, Astroparticle Physics, Quantum Field Theory, Neutrino Physics, Neutrino Astronomy
Programming: Computational Physics and Numerical Methods, Methods of Data Analysis

SKILLS

Languages: Python, C/C++, Bash, SQL
Frameworks & Tools: ROOT, SWIM, OscProb, PyTorch, GraphNet, TensorFlow, nuSQuIDS
Computing Methods: High-Performance Computing (SLURM, UNIX), Workflow Management (Snakemake), Version Control (Git), CI/CD (GitHub Actions), Containerization (Docker, Apptainer)
Foreign Languages: Greek (Native), English (C2), German (B2), Italian (A1)
Soft Skills: Team player, analytical thinker, problem-solving, adaptability

CONFERENCES & WORKSHOPS

Analysis Reproducibility <i>Participant</i>	Mar 2025 <i>Online</i>
12th HEP C++ Course and Hands-on Training - The Essentials <i>Participant</i>	Mar 2025 <i>Online</i>
KM3NeT Bootcamp <i>Participant</i>	Dec 2024 <i>Erlangen, DE</i>
DPG Spring Meeting <i>Speaker: Feasibility study of tau appearance measurement with the ANTARES neutrino telescope (T 29.2)</i>	Mar 2024 <i>Karlsruhe, DE</i>
KM3NeT Oscillation WG Meeting <i>Speaker: Feasibility study of tau appearance measurement with the ANTARES neutrino telescope</i>	Dec 2023 <i>Erlangen, DE</i>
KM3NeT Bootcamp <i>Participant</i>	Oct 2023 <i>Online</i>
GraphNeT III - Workshop <i>Participant</i>	May 2023 <i>Sandvig, DK</i>

VOLUNTEERING

Long Night of Sciences, FAU <i>Volunteer - Neutrino Group ECAP</i> Presented neutrino detection concepts to 1100+ visitors at ECAP through posters and demonstrations, as well as showcasing the group's contributions to neutrino physics & astronomy.	Erlangen, Germany <i>October 2023</i>
AUTh - CERN Masterclass <i>Volunteer</i> Guided 10 high school students through ATLAS event identification exercises during the AUTh-CERN Masterclass.	Thessaloniki, Greece <i>Mar 2021</i>
Physicists of Aristotle University of Thessaloniki (PATH) <i>Group Coordinator</i> - Organized the speaker list and conference logistics to highlight undergraduate research in physics for the second iteration of the $\Sigma\Pi E\Phi$ conference. - Facilitated student visits to research labs through the 'Labs & Paths' event, helping fellow students engage with the research conducted in the department.	Thessaloniki, Greece <i>Nov 2017 – Jun 2019</i>

REFERENCES

PD Dr. Thomas Eberl thomas.eberl@fau.de	Privatdozent, Erlangen Centre for Astroparticle Physics & FAU Erlangen ANTARES & KM3NeT Collaboration
Dr. Ekaterini Tzamariudaki katerina@inp.demokritos.gr	Lead Researcher, National Centre for Scientific Research Demokritos KM3NeT Collaboration
Dr. Rodrigo Gracia-Ruiz rgracia@km3net.de	Postdoctoral Researcher, Erlangen Centre for Astroparticle Physics ANTARES & KM3NeT Collaboration

PUBLICATIONS

1. Aiello, S. *et al.* gSeaGen code by KM3NeT: An efficient tool to propagate muons simulated with CORSIKA. *Comput. Phys. Commun.* **314**, 109660. arXiv: 2410.24115 [hep-ex] (2025).
2. Aiello, S. *et al.* Measurement of neutrino oscillation parameters with the first six detection units of KM3NeT/ORCA. *JHEP* **10**, 206. arXiv: 2408.07015 [hep-ex] (2024).
3. Aiello, S. *et al.* Search for neutrino emission from GRB 221009A using the KM3NeT ARCA and ORCA detectors. *JCAP* **08**, 006. arXiv: 2404.05354 [astro-ph.HE] (2024).
4. Aiello, S. *et al.* Atmospheric muons measured with the KM3NeT detectors in comparison with updated numeric predictions. *Eur. Phys. J. C* **84**, 696. arXiv: 2403.11946 [astro-ph.HE] (2024).
5. Aiello, S. *et al.* Differential Sensitivity of the KM3NeT/ARCA detector to a diffuse neutrino flux and to point-like source emission: Exploring the case of the Starburst Galaxies. *Astropart. Phys.* **162**, 102990. arXiv: 2402.09088 [astro-ph.HE] (2024).
6. Aiello, S. *et al.* Astronomy potential of KM3NeT/ARCA. *Eur. Phys. J. C* **84**, 885. arXiv: 2402.08363 [astro-ph.HE] (2024).
7. Aiello, S. *et al.* Searches for neutrino counterparts of gravitational waves from the LIGO/Virgo third observing run with KM3NeT. *JCAP* **04**, 026. arXiv: 2311.03804 [astro-ph.HE] (2024).
8. Unbehaun, T. *et al.* Prospects for combined analyses of hadronic emission from γ -ray sources in the Milky Way with CTA and KM3NeT. *Eur. Phys. J. C* **84**, 112. arXiv: 2309.03007 [astro-ph.HE] (2024).
9. Aiello, S. *et al.* Embedded software of the KM3NeT central logic board. *Comput. Phys. Commun.* **296**, 109036. arXiv: 2308.01032 [astro-ph.IM] (2024).